



ANSWER SCHEME DAQ SES DP014 2024/2025_SET B

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NO.	ANSWER SCHEME	MA															
1. a	$0.795 \text{ g cm}^{-3} \rightarrow \text{kg m}^{-3}$ $= \frac{0.795 \text{ g}}{\text{cm}^3} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{100^3 \text{ cm}^3}{1 \text{ m}^3}$ $= 795 \text{ kg m}^{-3}$																
1. b	<table border="1"> <thead> <tr> <th>Vector (km)</th><th>S_x (km)</th><th>S_y (km)</th></tr> </thead> <tbody> <tr> <td>2.50</td><td>2.50</td><td>0</td></tr> <tr> <td>4.00</td><td>$4.00 \cos 45^\circ = 2.83$</td><td>$4.00 \sin 45^\circ = 2.83$</td></tr> <tr> <td>S</td><td>S_x</td><td>S_y</td></tr> <tr> <td>Total</td><td>0</td><td>3.80</td></tr> </tbody> </table> x-comp: $\sum s_x = 0$ $0 = 2.50 + 2.83 + S_x$ $S_x = -5.33 \text{ km}$ y-comp: $\sum s_y = 0$ $3.80 = 0 + 2.83 + S_y$ $S_y = 0.97 \text{ km}$ $S = \sqrt{s_x^2 + s_y^2}$ $= \sqrt{(-5.33)^2 + (0.97)^2}$ $= 5.42 \text{ km}$ $\tan \theta = \frac{s_y}{s_x}$	Vector (km)	S_x (km)	S_y (km)	2.50	2.50	0	4.00	$4.00 \cos 45^\circ = 2.83$	$4.00 \sin 45^\circ = 2.83$	S	S_x	S_y	Total	0	3.80	
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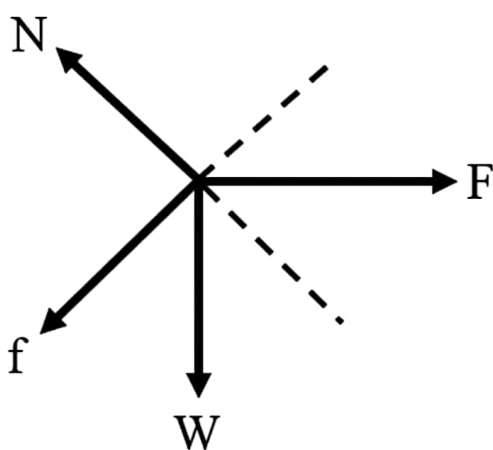
	$= \frac{0.97}{-5.33}$ $\theta = 10.3^\circ \text{ above } -x \text{ axis}$	
		TOTAL

NO.			ANSWER SCHEME	
2	a	i	The velocity of the object for the motions given b, BC and CD $v_{AB} = \frac{40 - 0}{4 - 0}$ $v_{AB} = 10 \text{ m s}^{-1}$ $v_{BC} = 0 \text{ m s}^{-1}$ $v_{CD} = \frac{0 - 40}{10 - 8}$ $v_{CD} = -20 \text{ m s}^{-1}$	2 / 12 – ⋮ G. J G.
		ii	The total distance moved $s = 40 + 40$ $= 80 \text{ m}$	.
		iii	The average speed of the object $v_{ave} = \frac{\text{distance}}{\text{time}} = \frac{80}{10}$ $v_{ave} = 8 \text{ m s}^{-1}$	(J
	b	i	$v^2 = u^2 - 2gs$ $v = 0 \text{ m s}^{-1} \text{ at } H_{max}$ $(0)^2 = (8)^2 - 2(9.81)H$	I (

			$H = 3.26 \text{ m}$	
		ii	$v^2 = u^2 - 2gs$ $v^2 = (0)^2 - 2(9.81)(-3.26)$ $v = 8 \text{ m s}^{-1}$	
		iii	$v = u - gt$ $-8 = 8 - (9.81)t$ $t = 1.63 \text{ s}$	
			TOTAL	13

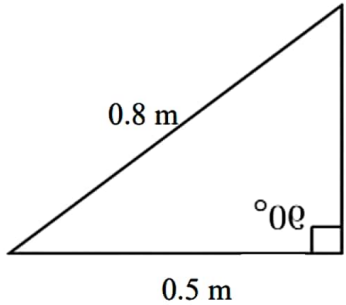
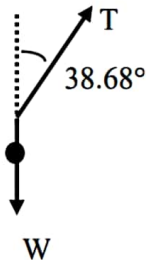
NO.		ANSWER SCHEME	3 / 12	⋮
3	a) i)	Impulse = Change of momentum $J = m(v - u)$ $J = (0.40)(18 - (-18))$ $J = 14.4 \text{ N s}$		
	a) ii)	$J = Ft = 14.4 \text{ N s}$ $14.4 = F(0.02)$ $F = 720 \text{ N}$		
	b)	$\sum p_i = \sum p_f$ For x-component, $m_A u_A + m_B u_B = m_A v_{Ax} + m_B v_{Bx}$ $m(30) + m(-15) = m(v_A \cos 30^\circ) + m(-v_B \cos 60^\circ)$ $15 = 0.866v_A - 0.5v_B \quad \dots(1)$ For y-component,		

		$m_A u_A + m_B u_B = m_A v_{Ay} + m_B v_{By}$ $0 = m(-v_A \sin 30^\circ) + m(v_B \sin 60^\circ)$ $0.5v_A = 0.866v_B$ $v_A = 1.732v_B \quad \dots (2)$ Substitute (2) into (1): $15 = 0.866(1.732v_B) - 0.5v_B$ $v_B = 15 \text{ m s}^{-1} \quad \dots (3)$ Substitute (3) into (2): $v_A = 1.732(15) = 25.98 \text{ m s}^{-1}$	
		Total	

NO.	ANSWER SCHEME	MARK
4(a) i		All for correct 3 force – D2 2 force – D1

4 (a) ii	$\sum F_y = 0$ $N - F \sin 30 - W \cos 30 = 0$ $N = 45 \sin 30 + 4(9.81) \cos 30$ $N = 56.48 N$	
4 (a) iii	$\sum F_x = ma$ $F \cos 30 - f - W \sin 30 = ma$ $45 \cos 30 - 0.25N - 4(9.81) \sin 30 = 4a$ $a = 1.31 m s^{-2}$	
4 (b)i	$W - T = 0$ $T = W = (5000)(9.81)$ $T = 49050 N$	5 / 12
4 (b) ii	$W - T = ma$ $T = W - ma$ $T = (5000)(9.81) - 5000(1.5)$ $T = 41550 N$	

NO.	ANSWER SCHEME	
5 (a) i.	$mass = 25 \text{ kg}$ $radius = 3.5 \text{ m}$ $\theta = 5 \text{ rev} = 10\pi \text{ rad}$ $t = 6 \text{ minutes} = 360 \text{ s}$ $\omega = \frac{\theta}{t} = \frac{10\pi}{360}$ $\omega = 0.087 \text{ rad s}^{-1}$	
5 (a) ii)	$F_c = ma_c = mr\omega^2$	

	$F_c = (25)(3.5)(0.087)^2$ $F_c = 0.662 \text{ N}$	
5 (b) i)	 $\sin \theta = \frac{r}{L}$ $\sin \theta = \frac{0.5}{0.8}$ $\theta = 38.68^\circ$	
(b) ii)	<i>mass</i> = 0.085 kg  $\Sigma F_y = 0$ $\Sigma F_y = T \cos \theta - W$ $T \cos 38.68^\circ - (0.085)(9.81) = 0$ $T = 1.068 \text{ N}$	6 / 12
	$\Sigma F_x = T \sin \theta = F_c$ $F_c = T \sin 38.68^\circ = \frac{m v^2}{r}$	

	$(1.068) \sin 38.68^\circ = \frac{(0.085) v^2}{0.5}$ $v = 1.981 \text{ m s}^{-1}$	G1 JU1
	TOTAL	9 M

NO.		ANSWER SCHEME	
6.	(a)	i) Given : $r = 0.05 \text{ m}$ $\omega = 250 \text{ rev min}^{-1}$ $v_t = ?$ $\omega = \frac{250 \times 2\pi}{60} = 26.18 \text{ rad s}^{-1}$ $v_t = r\omega$ $v_t = (0.05)(26.18)$ $\therefore v_t = 1.31 \text{ m s}^{-1}$	
		ii) Given : $t = 8.0 \text{ s}$ $\omega_0 = 0 \text{ rad s}^{-1}$ $\alpha = ?$ $\omega = \omega_0 + \alpha t$ $26.18 = 0 + (\alpha)(8.0)$ $\therefore \alpha = 3.27 \text{ rad s}^{-2}$	7 / 12 - 

	iii) Given : $t = 8.0 \text{ s}$ $\omega_0 = 0 \text{ rad s}^{-1}$ $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$ $\theta = (0)(8.0) + \frac{1}{2} (3.27)(8.0)^2$ $\therefore \theta = 104.64 \text{ rad}$ OR $\omega^2 = \omega_0^2 + 2\alpha\theta$ $\theta = \frac{\omega^2 - \omega_0^2}{2\alpha} = \frac{26.18^2 - 0^2}{2(3.27)}$ $\therefore \theta = 104.80 \text{ rad}$	(J (J
(b)	i) Given : $F = 5.0 \text{ N}$ $r = 2.6 \text{ m}$ $\theta = 33^\circ$  FIGURE 6.1 $\tau = r \times F$ $\tau = Fr \sin \theta$ $\tau = (5.0)(2.6) \sin 33^\circ$ $\therefore \tau = 7.08 \text{ Nm}$ The direction of torque is anticlockwise.	8 / 12 (J (

ii) Given :

$$F_1 = 30 \text{ N}, F_2 = 45 \text{ N}$$

$$d = 4.0 \text{ cm} = 0.04 \text{ m}, r = 0.02 \text{ m}$$

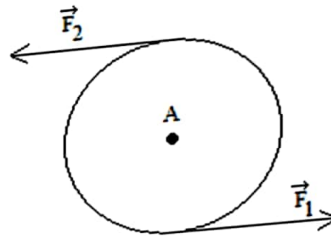


FIGURE 6.2

$$\sum \tau = \tau_{F1} + \tau_{F2}$$

$$\sum \tau = (30)(0.02)\sin 90^\circ + (45)(0.02)\sin 90^\circ$$

$$\therefore \sum \tau = 1.5 \text{ Nm}$$

$\therefore$ The direction is **anticlockwise**

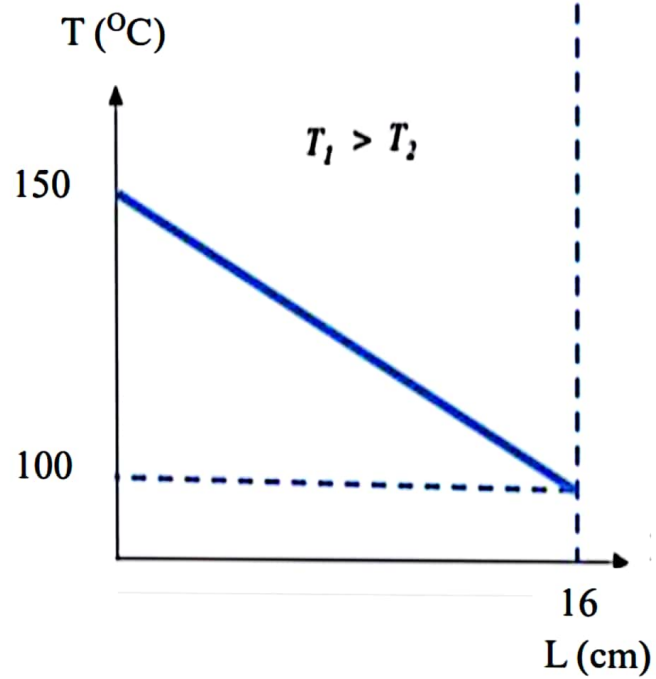
TOTAL MARKS

NO.	ANSWER SCHEME	MA
7. (a)(i)	$L = 5 \text{ m}$ $A = d^2 = (5)^2 = 25 \text{ m}^2$ $\Delta T = T_2 - T_1 = 0 - 60 = -60^\circ \text{C}$ $k = 209 \text{ W m}^{-1} ^\circ \text{C}^{-1}$ $t = 2 \text{ s}$ $\frac{Q}{t} = -kA \frac{\Delta T}{L}$ $Q = -209(25) \frac{-60}{5} (2)$ $Q = 1.254 \times 10^5 \text{ J}$	9 / 12 C Jl

(ii)	$k = 45 \text{ W m}^{-1} \text{ K}^{-1}$ $L = 10 \text{ cm} = 0.1 \text{ m}$ $\Delta T = -10 \text{ K}$ $Q/t = 50 \text{ J s}^{-1}$ $\frac{Q}{t} = -kA \frac{\Delta T}{L}$ $50 = -45(A) \frac{-10}{0.1}$ $A = 0.011 \text{ m}^2$	
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(iii)



$$L = 0.16 \text{ m}$$

$$T_2 = 100 ^{\circ}\text{C}$$

$$T_1 = 150 ^{\circ}\text{C}$$

$$\frac{\Delta T}{L} = \frac{T_2 - T_1}{L}$$

$$= \frac{100 - 150}{0.16}$$

$$= -312.5 ^{\circ}\text{C m}^{-1}$$

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(b)(i)	$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$ $M \text{ for CO}_2 = 44 \text{ g mol}^{-1}$ $T = 350 \text{ K}$ $v_{rms} = \sqrt{\frac{3RT}{M}}$ $= \sqrt{\frac{3(8.31)(350)}{0.044}}$ $= 445.32 \text{ m s}^{-1}$	
(ii)	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$ $T = 20 + 273.15 = 293.15 \text{ K}$ $v_{rms} = \sqrt{\frac{3kT}{m}}$ $511 = \sqrt{\frac{3(1.38 \times 10^{-23})(293.15)}{m}}$ $m = 4.65 \times 10^{-26} \text{ kg}$	
(c)(i)	$\Delta U = Q - W$ $30 = Q - 6$ $Q = 36 \text{ J}$	
(ii)	$W = \text{area under the graph}$ $W = P(V_2 - V_1)$ $W = 150 \times 10^3 (1 - 4)$ $W = -450000 \text{ J}$	
TOTAL		

